NCH-POUND

MIL-DTL-3928/18D
15 February 2011
SUPERSEDING
MIL-DTL-3928/18C
15 August 2001

DETAIL SPECIFICATION SHEET

SWITCHES, RADIO-FREQUENCY
TRANSMISSION LINE (COAXIAL) (ELECTRICALLY OPERATED)
CLASS 5, 1P6T

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

Requirements for acquiring the switch described herein
shall consist of this specification sheet and MIL-DTL-3928.

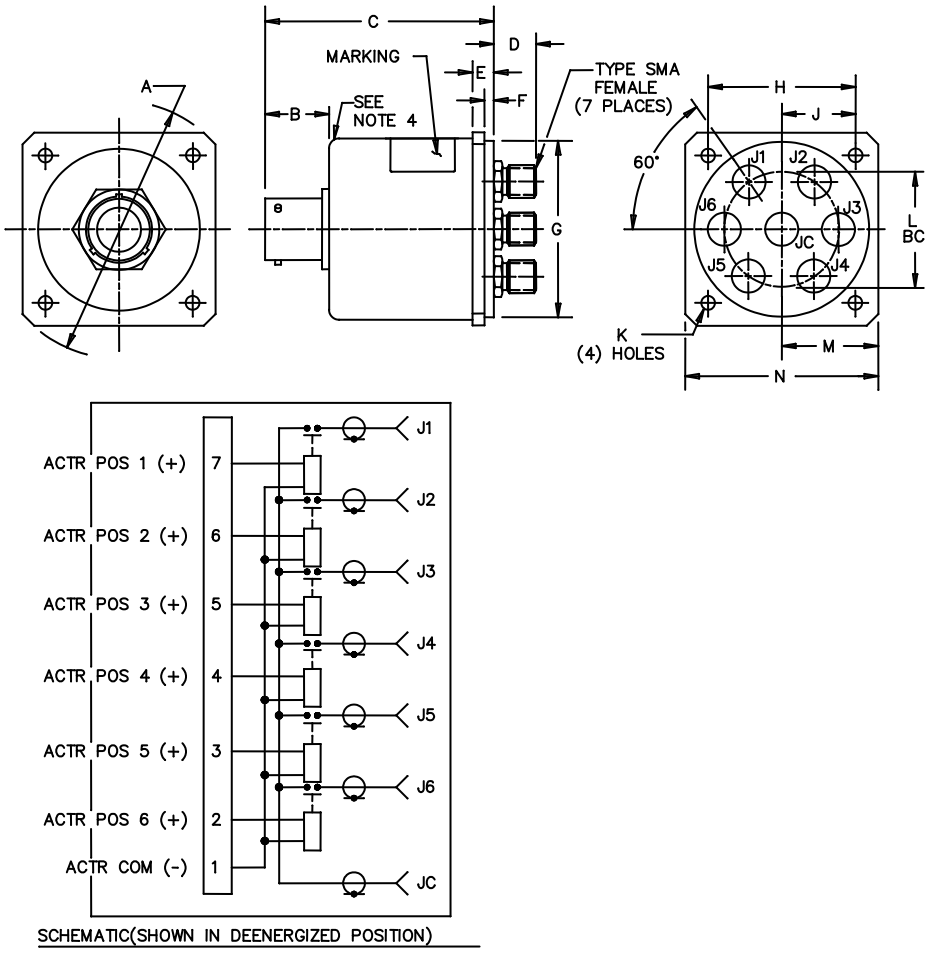


FIGURE 1. Switch configuration and schematic, PIN M3928/18-01.

AMSC N/A

FSC 5985

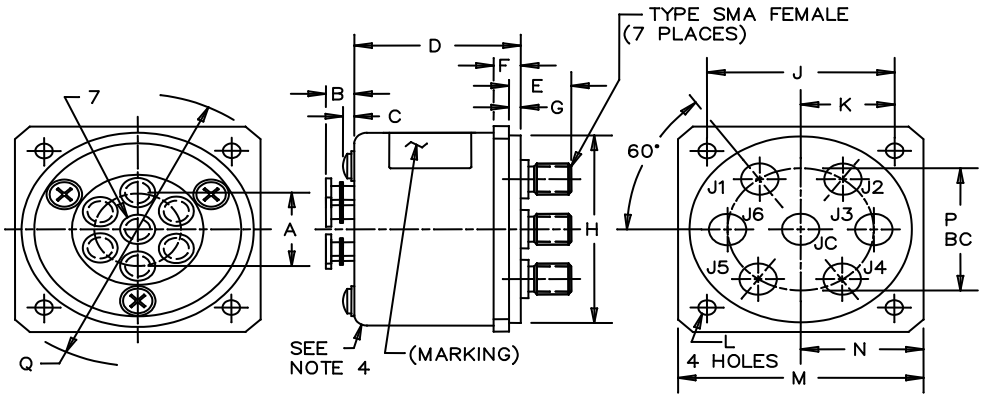
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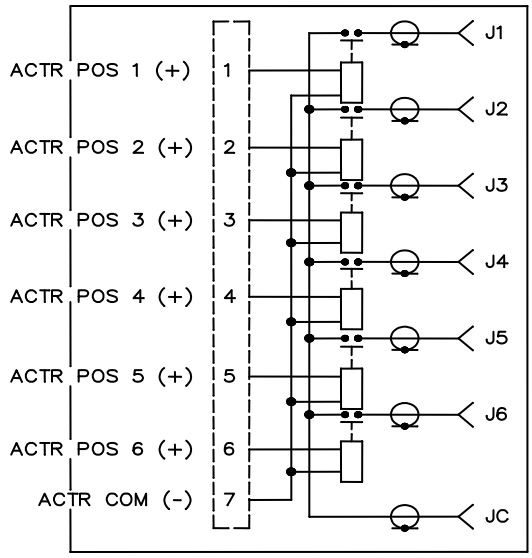
Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	2.34	2.28	59.4	57.9	H	1.380	1.370	35.05	34.80
B	.62	.56	15.7	14.2	J	.697	.677	17.70	17.20
C	1.60	---	40.6	---	K	.154 dia	.139 dia	3.91 dia	3.53 dia
D	.34	.28	8.6	7.1	L	1.072 dia	1.052 dia	27.23 dia	26.72 dia
E	.260	.240	6.60	6.10	M	.885	.865	22.48	21.97
F	.135	.115	3.43	2.92	N	1.78	1.72	45.21	43.69
G	1.560	1.556	39.62	39.52					

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 1. Switch configuration and schematic, PIN M3928/18-01 - Continued.

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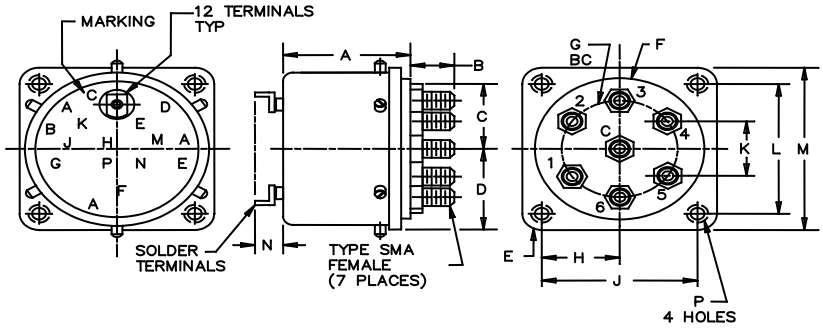
SCHEMATIC(SHOWN IN DEENERGIZED POSITION)

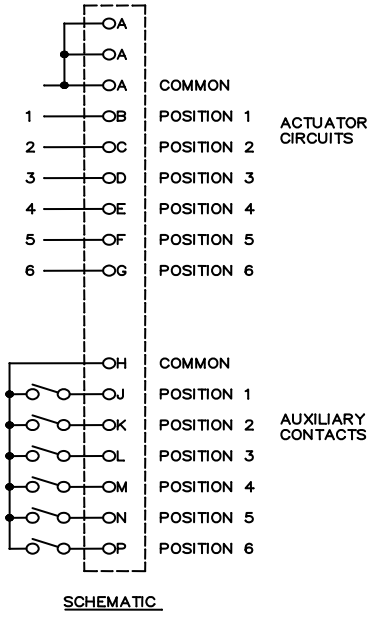
Dimensions									
	Inches		Millimeters			Inches		Millimeters	
Letter	Max	Min	Max	Min	Letter	Max	Min	Max	Min
A	.58 (BC)	.52 (BC)	14.7 (BC)	13.2 (BC)	J	1.380	1.370	35.05	34.80
B	.21	.15	5.3	3.8	K	.697	.677	17.70	17.20
C	.10	---	2.5	---	L	.154 dia	.139 dia	3.91dia	3.53 dia
D	1.55	---	39.4	---	M	1.78	1.72	45.2	43.7
E	.34	.28	8.6	7.1	N	.885	.865	22.48	21.97
F	.28	.22	7.1	5.6	P	1.072 dia	1.052 dia	27.23 dia	26.72 dia
G	.130	.120	3.30	3.05	Q	2.33 dia	2.27 dia	59.2 dia	57.7 dia
H	1.560	1.556	39.62	39.52					

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 2. Switch configuration and schematic, PINs M3928/18-02 and M3928/18-08.

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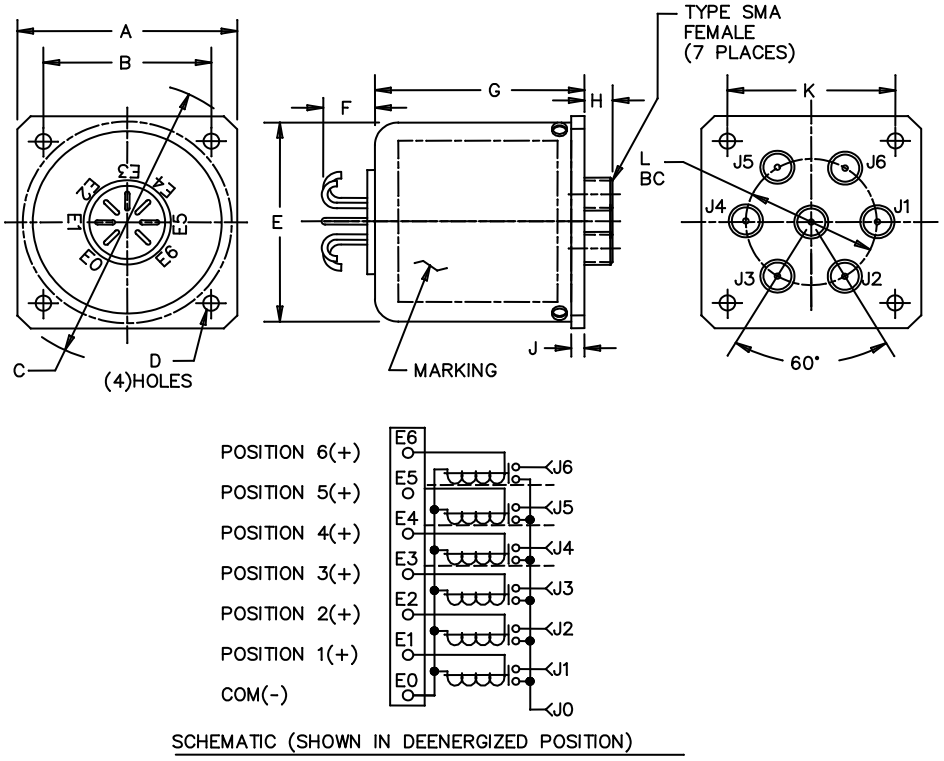
SCHEMATIC

Dimensions									
	Inches		Millimeters			Inches		Millimeters	
Letter	Max	Min	Max	Min	Letter	Max	Min	Max	Min
A	2.08	2.04	52.8	51.8	H	.692	.682	17.58	17.32
B	.33	.29	8.4	7.4	J	1.380	1.370	35.05	34.80
C	.692	.682	17.58	17.32	K	.55	.51	13.97	12.95
D	.880	.870	22.35	22.10	L	1.380	1.370	35.05	34.80
E	.20 R	.16 R	5.1 R	4.1 R	M	1.77	1.73	45.0	43.9
F	1.565 dia	1.555 dia	39.75 dia	39.50 dia	N	.35	.31	8.9	7.9
G	1.067 dia	1.057 dia	27.10 dia	26.85 dia	P	4-40 UNC-2B		.112-40 UNC-2B	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Metric equivalents are in parentheses.
4. Round corners of case may be squared.
5. Unless otherwise specified, tolerances are ± .005 (± 0.13 mm) for three place decimals and ± .02 (±0.5 mm) for two place decimals.

FIGURE 3. Outline and schematic drawing, PIN M3928/18-03.

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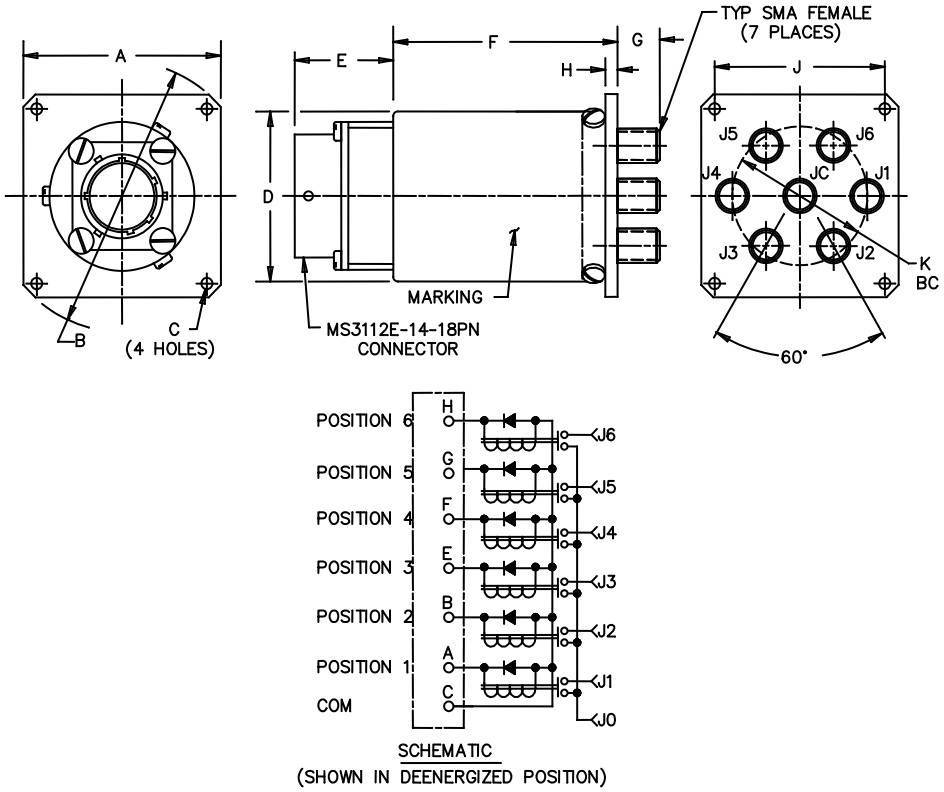
SCHEMATIC (SHOWN IN DEENERGIZED POSITION)

Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	1.77	1.73	45.0	43.9	G	1.85	---	47.0	---
B	1.65	---	41.9	---	H	.38	---	9.7	---
C	2.33	2.29	59.2	58.2	J	.130	.120	3.30	3.05
D	.135	.120 dia	3.43 dia	3.05 dia	K	1.380	1.370	35.05	34.80
E	1.52 dia	1.48 dia	38.6 dia	37.6 dia	L	.885 dia	.875 dia	22.48 dia	22.23 dia
F	.27	.23	6.9	5.8					

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .005 (± 0.13 mm) for three place decimals and ± .02 (±0.5 mm) for two place decimals.

FIGURE 4. Switch configuration and schematic, PIN M3928/18-04.

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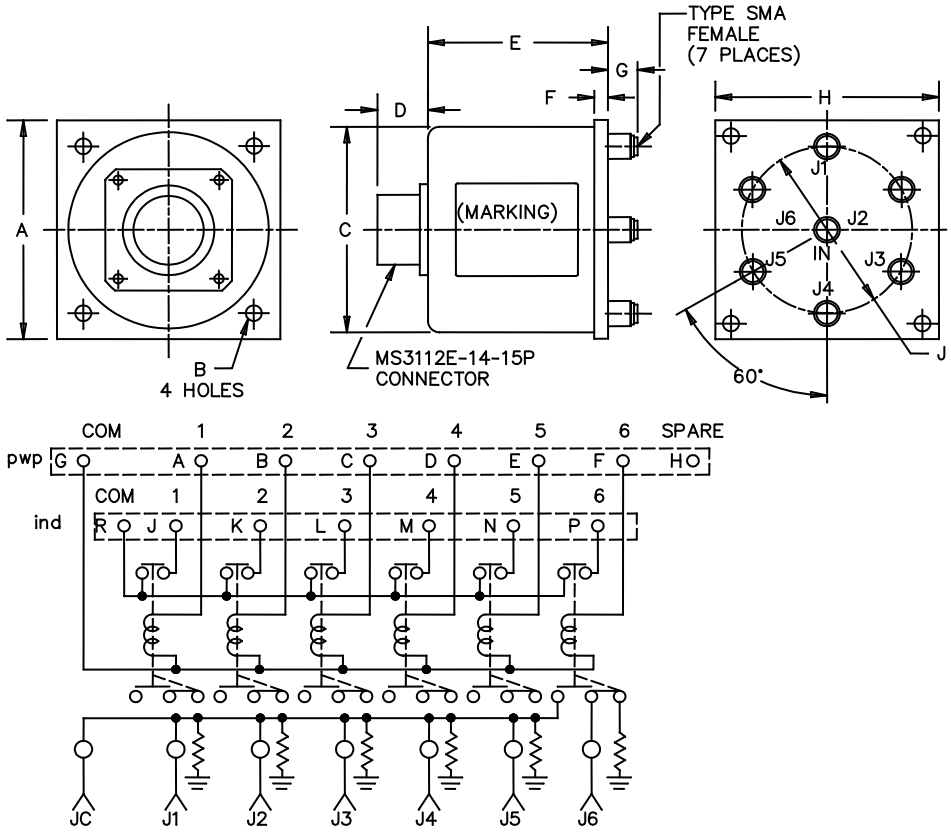


Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	1.77	1.73	45.0	43.9	F	1.85	---	47.0	---
B	2.33 dia	2.29 dia	59.2 dia	58.2 dia	G	.27	---	6.9	---
C	.135 dia	.120 dia	3.43 dia	3.05 dia	H	.130	.120	3.30	3.05
D	.152 dia	.148 dia	3.86 dia	3.76 dia	J	1.380	1.370	35.05	34.80
E	.95	---	24.1	---	K	.885	.875	22.48	22.23

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .005 (± 0.13 mm) for three place decimals and ± .02 (±0.5 mm) for two place decimals.

FIGURE 5. Switch configuration and schematic, PIN M3928/18-05.

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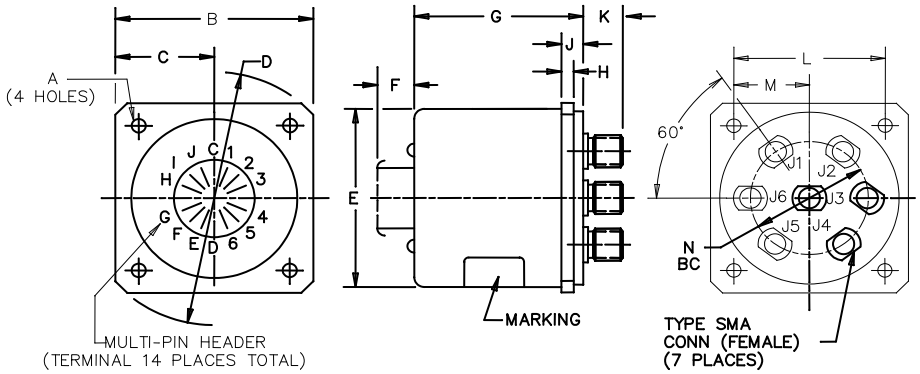


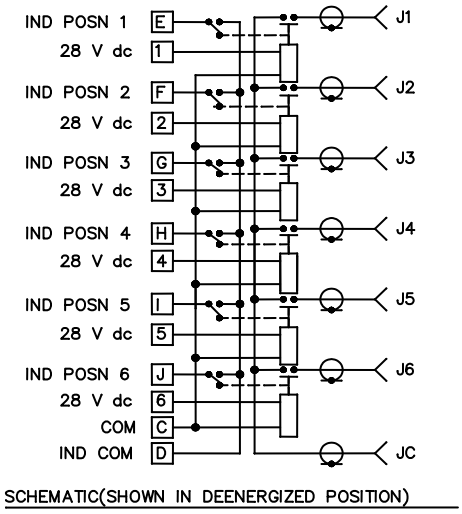
Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	2.28	2.22	57.9	56.4	F	.20	.14	5.1	3.6
B	.187 dia	.172 dia	4.75 dia	4.37 dia	G	.27	---	6.9	---
C	2.03 dia	1.97 dia	51.6 dia	50.0 dia	H	1.805	1.795	45.8	45.6
D	.53	.47	13.5	11.9	J	1.63 dia	1.57 dia	41.4 dia	39.9 dia
E	2.46	2.40	62.5	61.0					

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 6. Switch configuration and schematic, PIN M3928/18-06.

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SCHEMATIC(SHOWN IN DEENERGIZED POSITION)

Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	.154	.139	3.91	3.53	H	.135	.115	3.43	2.92
B	1.78	1.72	45.2	43.7	J	.28	.22	7.1	5.6
C	.885	.865	22.48	21.97	K	.31	---	7.9	---
D	2.34 dia	2.28 dia	59.4 dia	57.9 dia	L	1.385	1.365	35.18	34.67
E	1.568 dia	1.548 dia	39.83 dia	39.32 dia	M	.697	.677	17.70	17.20
F	.28	.22	7.1	5.6	N	1.072 dia	1.052 dia	27.23 dia	26.72 dia
G	2.00	---	50.8	---					

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 7. Switch configuration and schematic, PIN M3928/18-07.

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TABLE I. Electrical and performance characteristics.

PIN M3928/18-	Fig. No.	Housing	Frequency range dc to GHz	VSWR	Insertion loss (dB)	Isolation (dB)	Switch time (ms)	Position indication circuit and rating	Life cycles x 1000	Fail-safe or latching	Operating current (A) 1/	Holding current (A) 1/	Nominal operating voltage	Pickup voltage (less than)	Dropout voltage (less than)	Power and indicator connector	Weight (oz)
				Max	Max	Min	Max				Max	Max					Max
01N, S	1	I	18	1.5:1	0.5	60	20	None	1000	---	.17	.17	28 V dc	20 V dc	16 V dc	RTK07-8-7P (Deutsch or equal)	6
02N, S	2	I	18	1.5:1	0.5	60	20	None	1000	---	.17	.17	28 V dc	20 V dc	16 V dc	Solder terminals	6
03N, S	3	I	18 at 2-7 at 2-3	1.5:1 1.25:1	0.5 0.3	60 2/	50	None	1000	---	.22	.22	28 V dc	24 V dc	18 V dc 3/	Solder terminals	4.5
04N, S	4	I	18 at 6-12 at DC-6	1.5:1 1.4:1 1.25:1	0.5 0.4 0.25	60 60 70	20	None	1000	---	.17	.17	28 V dc	20 V dc	16 V dc	Solder terminals	6
05N, S 4/	5	I	18 at 6-12 at DC-6	1.5:1 1.4:1 1.25:1	0.5 0.4 0.2	60 60 70	20	None	1000	---	.16	.16	28 V dc	20 V dc	16 V dc	MS3112E14 -18PN	4
06N, S 5/	6	I	18	1.5:1	0.5	60	15	2A at 28 V dc	1000	---	.16	.16	28 V dc	20 V dc	16 V dc	MS3112E14 -15P	6
07N, S	7	I	18	1.5:1	0.5	60	20	2A at 28 V dc	1000	---	.18	.18	28 V dc	20 V dc	16 V dc	Solder terminals	6
08N, S	2	I	18.5	1.5:1	0.6	60	20	None	1000	---	.17	.17	28 V dc	20 V dc	16 V dc	Solder terminals	6

1/ At 28 V dc and 20 deg C.

2/ Isolation between connected ports and open ports shall be 6 dB minimum.

3/ Switch must return to deenergized position at no less than 2 volts.

4/ Contain ARC suppression diodes.

5/ Off-circuit output terminals are terminated with a 50 ohm, .5 watt resistor.

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REQUIREMENTS:

Dimensions and configurations: See figures 1 through 7.

Termination: Open, except for dash number 06, which is terminated.

Nominal impedance: 50 ohms.

RF connectors: Female connectors (7 places) shall meet the requirements of MIL-PRF-39012 and shall mate with SMA type male connectors in accordance with MIL-PRF-39012/55.

Electrical and performance characteristics: See table I.

Auxiliary contacts: 2 amps at 28 V dc for dash number 03.

RF power handling capability (average): 25 watts (minimum) except for dash number 03 which is 30 watts.

Transient interference (RFI): Required for M3928/18-03.

Vibration: Method I.

Operating temperature: -55 deg C to +85 deg C except for dash number 03 which is -55 deg C to +115 deg C.

Part or Identifying Number (PIN): M3928/18- (and dash number from table I).

Referenced documents: In addition to MIL-DTL-3928, this document references the following
MIL-PRF-39012
MIL-PRF-39012/55
MS3112

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5985-2011-009)

Review activities:
Army - MI
Navy - AS, CG, MC, OS
Air Force - 19, 99

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